

Evidence-Aware Hallucination Reduction for 3GPP RAN

Abhay Shashidhara, Chaitanya Kadam , Francisco Sevilla
Semantic Sages

Motivation

3GPP RAN specifications such as **TS 38.331** are long, technical, and require highly precise answers. Since language models can produce fluent but unsupported responses, this project focuses on building an **evidence-aware Q&A system** that grounds answers in retrieved specification text.

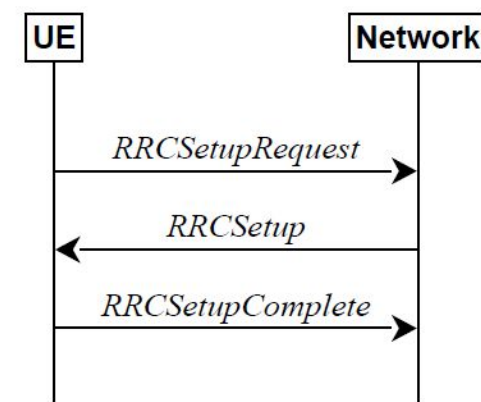


Figure 5.3.3.1-1: RRC connection establishment, successful



Literature Review

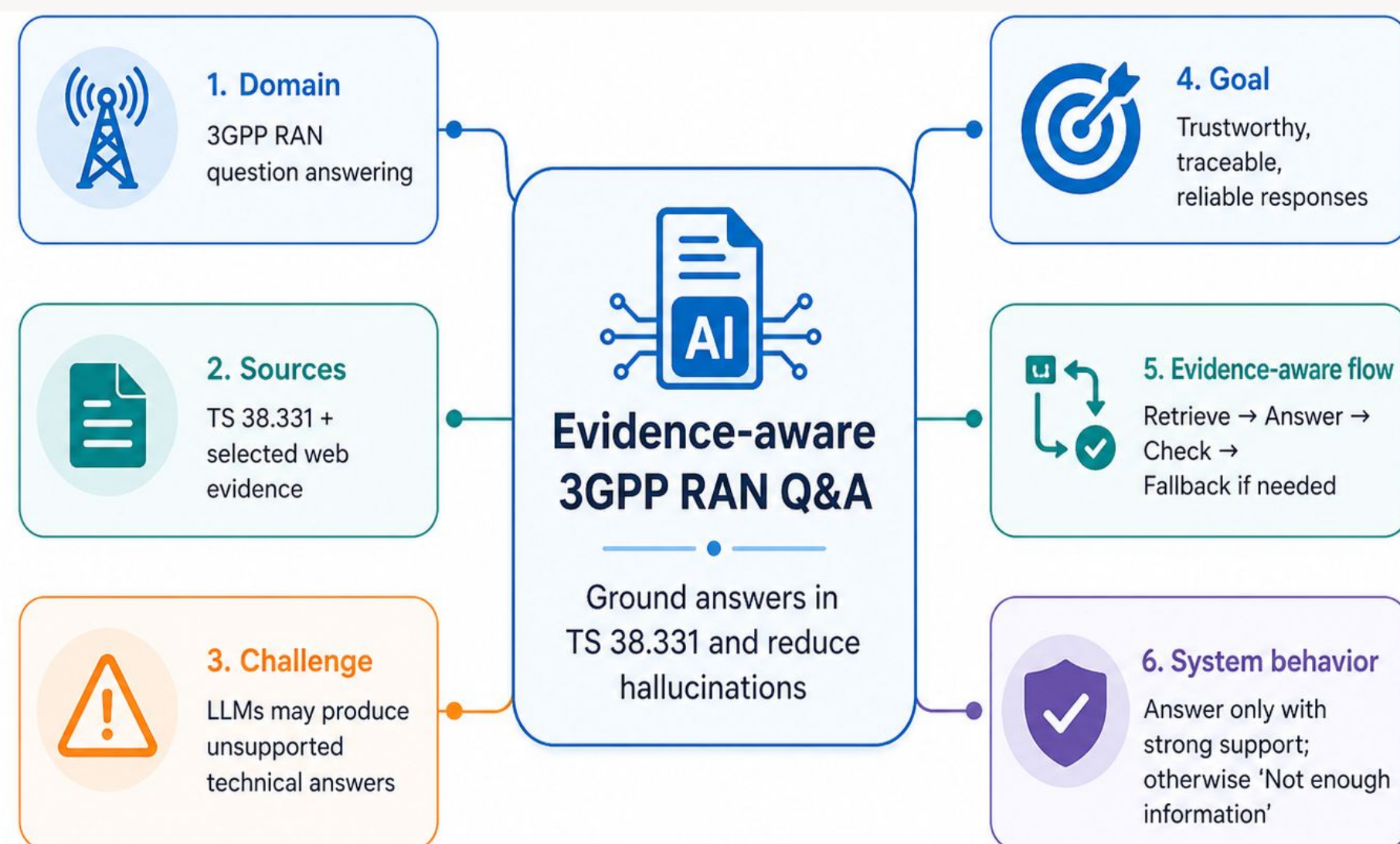
Prior work supports three parts of our design. **Lewis et al. [1]** motivates retrieval from **3GPP TS 38.331** instead of relying only on model knowledge. **Gao et al. [2]** supports making answers traceable to supporting evidence. **Asai et al. [3]** motivates our verification step, where an **NLI judge** checks whether the generated answer is supported.

Together, these works support our pipeline: **retrieve evidence, generate a grounded answer, and verify support** before returning the response.

Problem Definition

The problem is to design a Q&A assistant that can take technical 3GPP RAN questions and decide whether the available evidence is strong enough to answer.

- Input:** User questions about 3GPP RAN behavior, procedures, messages, and configuration details.
- Output:** A grounded technical answer when supporting evidence is available, or “**Not enough information**” when the evidence is weak.

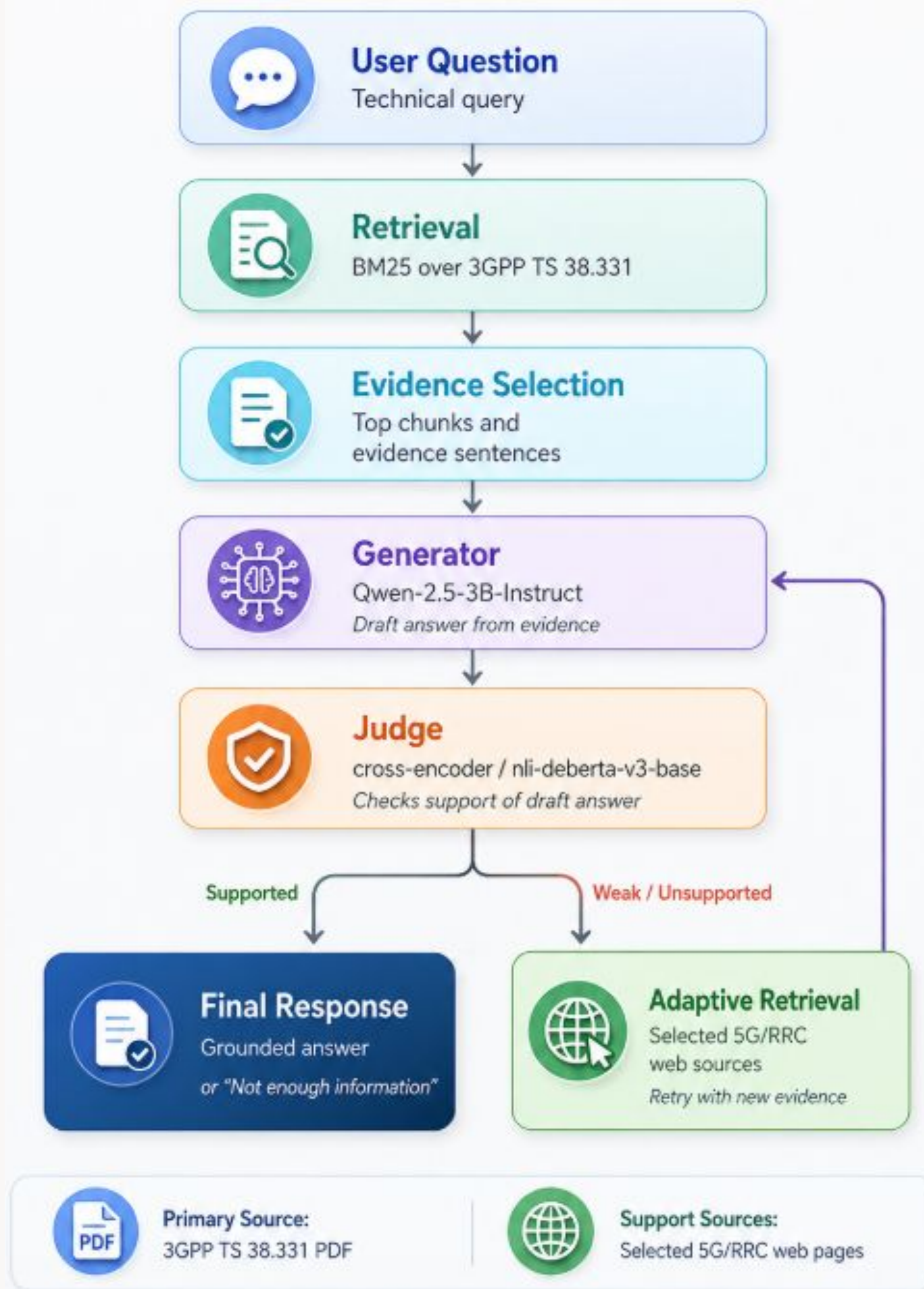


Datasets Used

The project uses **3GPP TS 38.331** as the primary specification source, supported by selected **5G/RRC web sources** for additional retrieval context.



Pipeline



- The pipeline starts with a **technical 3GPP RAN question** and retrieves evidence from **3GPP TS 38.331** using **BM25**.
- It uses **prefetch_k = 120**, keeps the final **top_k = 8** chunks, and selects up to **18 evidence sentences** with **minimum overlap = 0.01**.
- The generator, **Qwen-2.5-1.5B-Instruct**, creates a draft answer using **max_new_tokens = 450**.
- The **NLI judge** verifies the draft; if the primary answer is weak, the system uses **adaptive retrieval** with **adaptive_top_k = 8** over selected 5G/RRC sources.
- The adaptive stage gives the system a second chance to find better evidence before returning either a grounded answer or “**Not enough information.**”

Decision	Judge 1: Primary	Judge 2: Adaptive
Supported	Entailment ≥ 0.50 or support score ≥ 0.34	Entailment ≥ 0.35 or support score ≥ 0.20
Partially supported	$0.25 \leq \text{entailment} < 0.50$	$0.20 \leq \text{entailment} < 0.35$
Unsupported / Reject	Contradiction > 0.75 or below partial threshold	Contradiction > 0.80 or below partial threshold

Fig. 1: NLI Judge Decision Thresholds using cross-encoder/nli-deberta-v3-base.

Evaluation

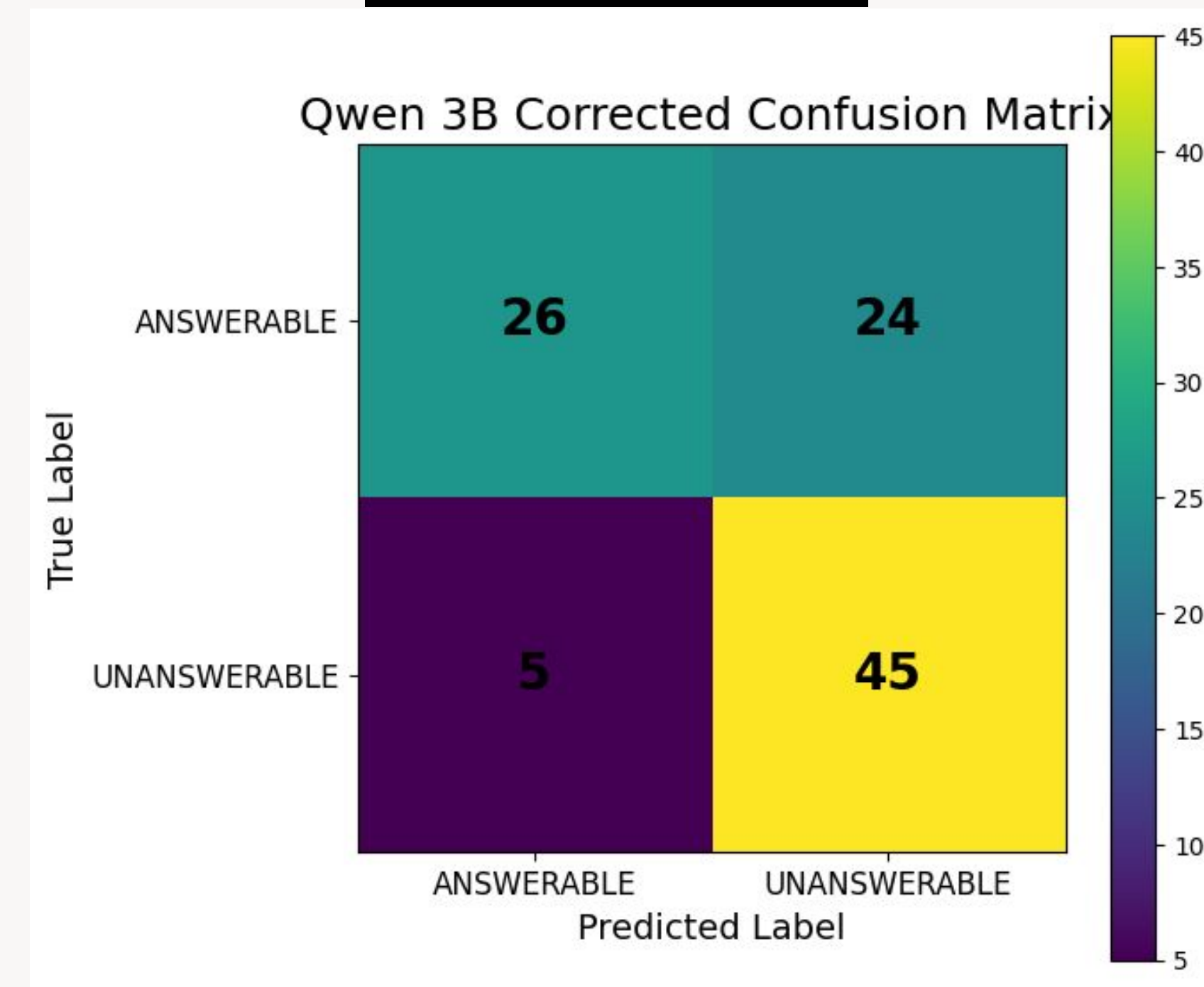


Fig. 2: Confusion Matrix for Qwen2.5-3B

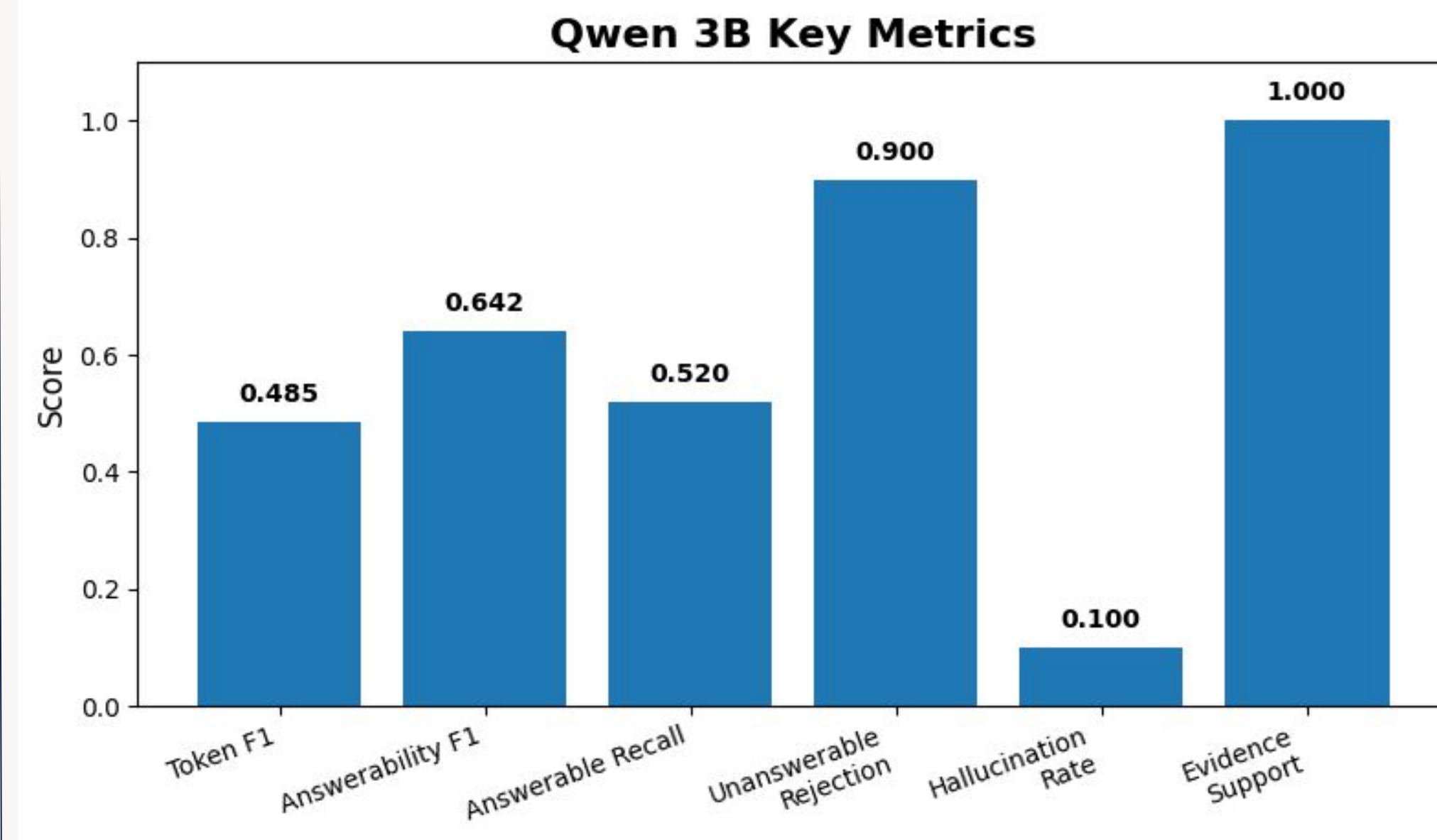


Fig. 3: Key Metrics for Qwen2.5-3B

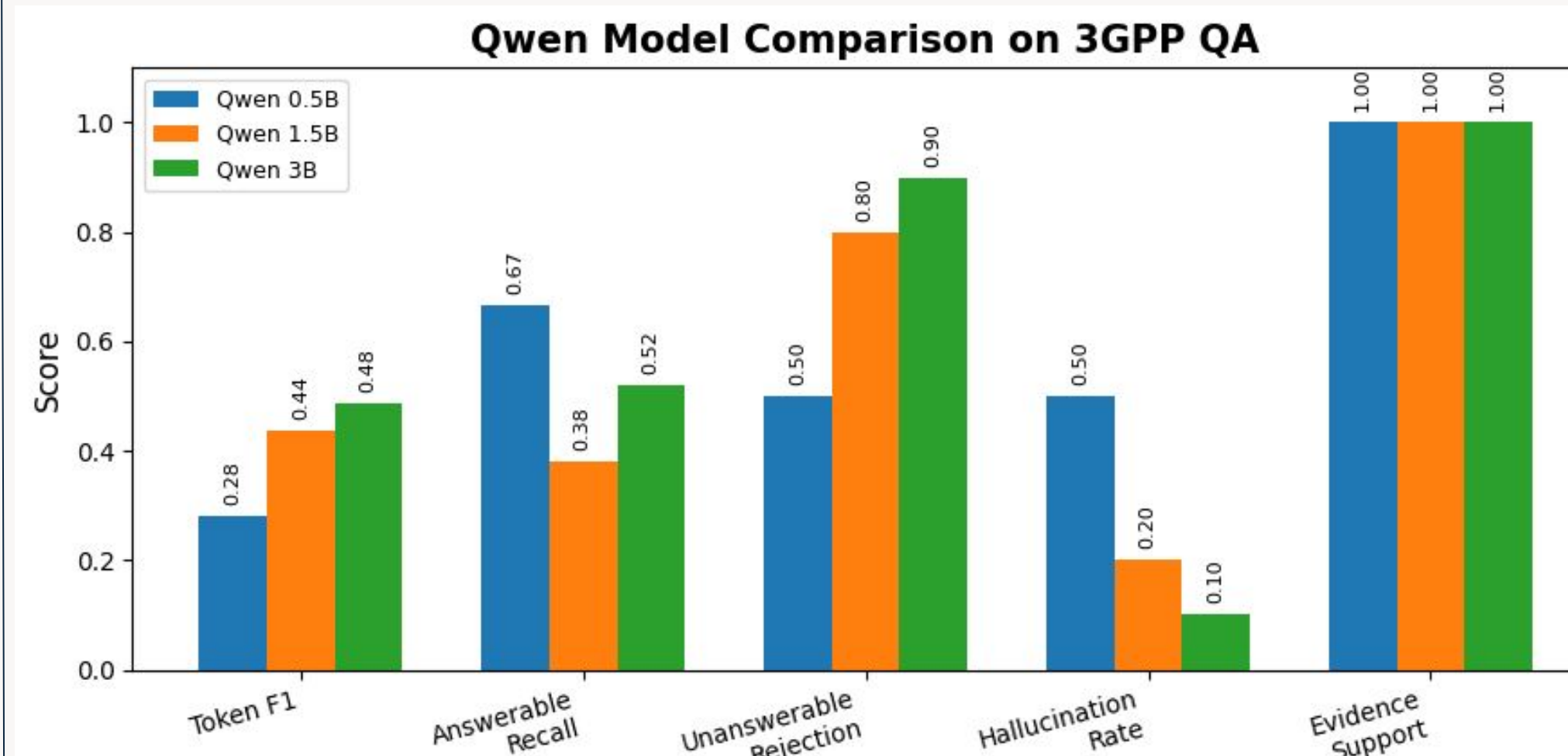


Fig. 4: Qwen Model Comparison on 3GPP QA

Conclusion

- Qwen 3B achieves the **highest unanswerable rejection rate (0.90)**, the **lowest hallucination rate (0.10)**, and the **best token F1 (0.485)**.
- It is **conservative, answering only 52%** of answerable questions.
- Qwen 3B **reduces unsafe answering**, The trade-off is **lower answer coverage**, since strict verification causes the system to abstain on some answerable 3GPP questions.

Plan for Final Report

In the final report, we will provide a comparative analysis of the models used and discuss how each configuration affects answer quality, speed, and hallucination reduction. We will also expand the findings with more experiments, detailed validation results, and observations on when the system performs well or fails.

References

- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. **Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks**. *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. **Enabling Large Language Models to Generate Text with Citations**. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. **Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection**. *International Conference on Learning Representations (ICLR)*, 2024.